

Mounir Lbath

mounirlbath.github.io | github.com/mounirLbath | linkedin

EDUCATION

Ecole Polytechnique

2024 – 2028

MSc, Cycle Ingenieur Polytechnicien; Applied Math + CS (AI specialization); GPA 4.19/4.30

Palaiseau, France

- Data Science, Algorithms and Data Structures, Statistics and Stochastic Processes, Optimization and Control, PDEs and Variational Analysis, Computer Graphics...

Lycee Louis-le-Grand

2022 – 2024

Preparatory Classes MPSI/MP, Intensive math, CS, and physics undergraduate program; GPA 4.0/4.0*

Paris, France

- Results: ranked 7th at Ecole Polytechnique, 7th at ENS Ulm nationwide

EXPERIENCE

Software Engineering Intern, Databricks

June 2026 – Aug. 2026

Object Storage

Berlin, Germany

- Shipped the move, rename, and copy file operations in the Files API for Unity Catalog (Scala, Protocol Buffers), on a surface serving 1M+ queries/day
- Ensured correctness under concurrent writes by removing race conditions and data-loss windows from the commit path; implemented the operations against the AWS, GCP, and Azure storage APIs

RESEARCH

Research Project, Ecole Polytechnique / INRIA

Sept. 2026 – Present

Geometric deep learning. Advisor: Dr. Maks Ovsjanikov

Palaiseau, France

- Studying attention layers as operators on function spaces, using spectral and geometric tools

Research Project, Ecole Polytechnique / INRIA SODA

Sept. 2025 – Apr. 2026

Knowledge tracing and adaptive learning. Advisor: Dr. Jill-Jenn Vie

Palaiseau, France

- First-author paper, AIED 2026: *Live Knowledge Tracing: Real-Time Adaptation using Tabular Foundation Models*
- Showed tabular foundation models trace knowledge by in-context learning alone, adapting to a learner at inference time with no gradient updates (achieving 53x speedups)
- Reimplemented attentive knowledge tracing from scratch in Pytorch and tested it against benchmarks

AWARDS & COMPETITIONS

- **Silver Medal**, International Physics Olympiad (2023)
- **Gold Medal**, International Mathematics Competition (2025); **Gold Medal**, French Mathematical Olympiad (2021)
- **ICPC SWERC 2025**: one of three teams representing Ecole Polytechnique; **Prologin finalist**: 21st nationwide

PROJECTS

Real-Time Gaussian Splatting Renderer

2026

C++, OpenGL/GLSL (repo)

- Built a real-time Gaussian splatting renderer (PLY parser, GPU instancing, projection, depth sorting) in C++/OpenGL, sustaining 300+ FPS
- Extended it with MILo mesh extraction driving a rigid-body simulator, enabling collisions between splat-rendered objects

Hackathon wins, Tech Europe (2 wins): education track win, an app teaching LLM-assisted coding (Next.js, Redis); viral content track win, a GitHub repo analyzer

2025

Representation Theory Memoir: a self-contained exposition of representation theory and its application to the proof of Frobenius' theorem in group theory ([link](#))

2024

LEADERSHIP

President, Binet IA, Ecole Polytechnique's AI association: ran hackathons, co-founder and chair of the Hi! Paris Student AI Conference (Nov. 2026), a peer-reviewed student venue with a senior academic committee

2026 – Present

Teaching for inmate rehabilitation: Robotics (Arduino), Math, Science, English

Oct. 2024 – Apr. 2025

Mathematics Instructor, Lyon Math Circle and French Olympiad Preparation

2022 – 2024

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, Scala, C#, OCaml, SQL

Tools: PyTorch, Git, Linux, Protocol Buffers, Next.js, OpenGL, Unity